Exp. 14 Implement code RAW Sockets
E.no.26 for packet sniffing.

Aim:

To write a code using RAW sockets to implement packet sniffing.

Code:-

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = "TCP"
        elif protocol == 6:
            protocol_name = "TCP"
        else:
            protocol_name = "Unknown Protocol"

        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
```

```
print("_" * 80)

sniff(iface = "wifi", prn = packet_callback,
      filter = "ip", store = 0)
```

Output :-

```
Protocal : TCP
Source IP : 57.155.141.117
Destination IP : 192.168.1.6

Protocal : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

Protocal : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1
```

Result :-

Thus the program code using RAW sockets to implement packet sniffing was executed successfully.